INCH-POUND

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28 September 1998
SUPERSEDING
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6 April 1993

PERFORMANCE SPECIFICATION

BAG, ODOR BARRIER, FLEXIBLE
(FOR FOOD CONTAMINATED PLASTIC WASTE)

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of odor barrier bags used to contain food-contaminated plastic waste (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in Sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8105

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC-Z1.4	-	Procedures, Sampling and Tables for Inspection by Attributes. (DoD Adopted)

(Applications for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 53021-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D882	-	Plastic Sheeting, Tensile Properties of Thin. (DoD adopted)
ASTM-D1922	-	Propagation Tear Resistance for Plastic Film and Thin Sheeting by Pendulum Method. (DoD adopted)
ASTM-D3985	-	Standard Test Method for Oxygen Gas Transmission Rate Through Plastic Film Sheeting Using a Kilometric Sensor.
ASTM-F88	-	Seal Strength of Flexible Barrier Materials, Test Method for
ASTM-F1249	-	Standard Test Method for Water-Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor.

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), samples shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. The film shall be made from such materials and by such processes as to ensure compliance with the requirements of this specification.

3.3 Form. The film shall be furnished in rolls formed of lay flat tubing.

3.3.1 Rolls. The average roll length shall be 75 ± 0.5 yards (64 ± 0.5 meters). The length of any individual roll shall be not less than 67 yards (61 meters). The width of roll material shall be 36 ± 0.25 inches (91 ± 0.6 centimeters) for size A bags or 24 ± 0.25 inch (61 ± 0.6 centimeters) for size B bags. The length of the core shall be equal to the width of roll material, with a plus

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tolerance of 0.25 inches (0.64 centimeters). The core shall be rigid to prevent distortion of the roll during use and shipment. Each roll shall be restrained to prevent unwinding (see 4.3.2).

3.3.1.1 Bags. Bags shall be created on the roll of lay flat tubing, by creating a heat-seal every 50 + 0.25 inches (127 + 0.64 centimeters) for the size A bag, or 27 ± .02 inch (68 ± 0.6 centimeters) for the size B bag. Bag sizes (see 6.2) shall be as specified in table I. Heat-seals shall meet the requirements specified in 3.9. Perforations shall be created to enable bags to be removed from the roll by hand. The perforations shall occur below the heat-seal, and shall be not more than 0.25 inches (0.64 centimeters) below the effective heat-seal. Perforations shall not interfere with the integrity of the seal (see 4.3.2).

TABLE I. Bag Sizes.

Size	Length (inches)	Width (inches)
A	50	36
B	27	24

3.4 Thickness. Unless otherwise specified (see 6.2), the film thickness shall be 0.005 inches (0.013 centimeters). When other thicknesses are required, they shall be specified in increments of 0.001 inches (0.025 centimeters). The tolerance shall be ± 5 percent of the specified thickness when measuring the average gage (see 4.3.2).

3.5 Solvent compatibility. The film shall be capable of meeting 75 percent of each mechanical property value specified in 3.7 and shall reveal no visual defects after being treated in accordance with 4.4.2 with any of the following solvents:

a. Chloroform
b. Xylene
c. Kerosene

3.6 Permeability. The water vapor permeability shall be not greater than 5.97 grams per square meter per day for 0.005 inch (0.013 centimeters) thick film material when tested in accordance with 4.4.1. The oxygen permeability shall be not greater than 1.16 cubic centimeters (standard temperature and pressure) per square meter per day per atmosphere for 0.005 inch (0.013 centimeters) thick film material, when tested in accordance with 4.4.1.

3.7 Tensile strength, tear strength and elongation. The film tensile strength shall be not less than 4,910 pounds per square inch (33.8 mega pascals) in the machine direction and 4,700 pounds per square inch (32.4 mega pascals) in the transverse direction, when tested in accordance with 4.4.3. The film elongation shall be not less than 280 percent in the machine and 294 percent in the transverse direction, when tested in accordance with 4.4.3. The film tear strength shall be not less then, 96 grams-force in the machine direction and 124 grams-force in the transverse direction, when tested in accordance with 4.4.3.

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3.8 Identification sheet. Each roll shall contain an identification sheet with the specification number, manufacturer's name, manufacturer's designation, lot number, date of manufacture (month and year), size, and sealing information; type of equipment, temperature, pressure, and dwell time (see 4.3.2).

3.9 Heat-seal strength. Heat-seals shall be 0.25 ± 0.05 inches (0.64 ± 0.13 centimeters). Seals shall be the flat impulse type whereby the film is heated to its welding or melting temperature under pressure and allowed to cool while still under pressure. Heat-seals shall have a breaking strength that shall be not less than 15 pounds per inch (263 newtons per meter), when tested in accordance with 4.4.4.

3.10 Workmanship. The film shall be free from any foreign matter. The film edges shall be cut and trimmed of any selvage. The material shall be free from holes, tears, cuts, sharp creases, wrinkles, or other imperfections (see 4.3.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of all the tests and examinations specified in this specification.

4.3 Conformance inspection. Conformance inspections consist of the required tests listed in table II and the examinations listed in table III.

TABLE II. Conformance tests.

Characteristics	Requirement paragraph	Test paragraph
Solvent compatibility	3.5	4.4.2
Permeability	3.6	4.4.1
Tensile strength	3.7	4.4.3
Tear strength	3.7	4.4.3
Elongation	3.7	4.4.3
Heat-seal strength	3.9	4.4.4

4.3.1 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in ASQC-Z1.4.

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4.3.2. Examination of end item for defects in appearance, construction, and workmanship. The end item shall be examined for the defects specified in table III.

TABLE III. Examination of end items for defects in appearance, construction and workmanship.

Examination	Defects
Form	Not rolled or flat cut as specified. Dimensions or film thickness not as specified. Core is distorted. Roll not restrained.
Bags	Heat-seal not as specified in 3.3.1.1. Bag sizes not specified in table I. Perforations missing or do not allow bags to be removed by hand. Perforations not located below heat-seal as specified in 3.3.1.1.
Appearance	Not clean.
Workmanship	Embrittlement or any holes, tears, cuts chafed spots, or other defects
Construction	Not uniform; pinholes, cracks, blister scratches, mottling, folds, foreign matters, gels or fish eyes. Edges not clean cut.
Color	Not uniform.
Identification	Marking not as specified; identification sheet missing, sealing instructions missing or incomplete.

4.4 Test methods. Unless otherwise specified all tests shall be conducted as specified herein. Film samples shall be conditioned for 24 hours (min) in a circulating air environment having a relative humidity of 50 ± 5 percent and a temperature of 70 to 76 ºF (21 to 25 ºC).

4.4.1 Permeability. The water-vapor permeability shall be measured in accordance with ASTM-F1249. The oxygen permeability shall be tested in accordance with ASTM-D3985. Oxygen permeability shall be tested at 0 percent relative humidity.

4.4.2 Solvent compatibility. Film samples shall be immersed in the solvents (a separate specimen for each solvent) specified in 3.5 for 7 days at 68 ± 5 ºF (20 ± 3 ºC), and a visual examination after treatment shall be performed. The treated film shall be tested in accordance with 4.4.3 to determine compliance with 3.5.

4.4.3 Film tensile strength, tear strength, and elongation. The film tensile strength and elongation shall be tested in accordance with ASTM-D882. The film tear strength shall be tested in accordance with ASTM-D1922 to meet requirements of 3.7.

4.4.4 Heat-seal strength. The film strength shall be tested in accordance with ASTM-F88.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points packaging activity within the Military Department or Defense Agency, or within The Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The bags covered by this specification are used in military shipboard applications and are exposed to prolonged periods of extreme seagoing environments not encountered by commercial bags. Flexible odor barrier bags in accordance with this specification are to be used with heat-sealing equipment. The bags provide for the containment of food-contaminated plastic waste in shipboard applications where at-sea disposal of plastics is prohibited.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Bag size (see table I).

d. Thickness (if other than 0.005 inches) (see 3.4).

e. Details for first article inspections/testing (see 4.2 and 6.3).

f. Packaging requirements (see 5.1).

6.3 First article. The contracting officer should provide specific guidance to offerors whether the first article samples should be a preproduction item, a first production item, a sample selected from the first production items or a standard production item from the contractor's current inventory (see 3.1 and 4.2). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for submission of Material Safety Data Sheets (MSDS) prior to contract award, testing, approval of first article test results, and

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disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive requirements for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government. Bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that the prior Government approval is acceptable for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.3.1 Prior approval. If a contractor has previously delivered an acceptable product meeting the requirements of this specification, waiver of the first article inspection requirement is at the discretion of the procuring activity for a period of time not to exceed 2 years (see 6.2).

6.3.2 First article sample. The first article sample should be produced using the same plant facilities intended for the production run. Failure of the first article sample to meet all the requirements of this specification is cause for rejection.

6.4 Lot size. An inspection lot should consist of 665 rolls of material, manufactured by the same process, from the same components, by one manufacturer, at one plant, and offered for delivery at one time.

6.4 Subject term (keyword) listing.

Heat-seal
Permeability
Plastic film
Oxygen gas transmission rate
Waste bag

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy - AS

Preparing Activity:
Navy - AS

(Project 8105-N002)

Review activity:
Navy - SA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. **The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.**

2. **The submitter of this form must complete blocks 4, 5, 6, and 7.**

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   NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-29597A(NAVY)	2. DOCUMENT DATE (*YYMMDD*) 980928

BAG, ODOR BARRIER, FLEXIBLE (FOR FOOD CONTAMINATED PLASTIC WASTE)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial: (2) DSN: (*If Applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial: (732) 323-2947 (2) DSN: 624-2947
c. ADDRESS (*Include Zip Code*) CODE 414100B120-3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Logistics Agency (DLSC-LM), Attn: Carla Jenkins/John Tascher 8725 John J. Kingman Road, Ste 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6874

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